

โครงสร้างการแบ่งงาน

Work Breakdown Structure

Phase 1 — Deliverable-oriented hierarchy (Level 1–4, 100% rule)

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ข้อมูลควบคุมเอกสาร

Control record for Work Breakdown Structure

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02 APPROVAL & OWNERSHIP การอนุมัติและความเป็นเจ้าของ

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03 LIFECYCLE วงจรชีวิตเอกสาร

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DATE / วันที่	REV	DESCRIPTION / รายละเอียดการแก้ไข	BY / โดย
27 July 2026	0.7	v0.7: แก้ consistency ทั้งหมด — SRS solo dev (เดิม 2 คน), REQ SCS→SCH + ปรับเลข WBS ให้ตรงรหัสจริง + เพิ่ม traceability S5 ครบ (เพิ่ม F-014/015, N-003/004/006/007/008/010, R-005) + SA Reference จาก SRS, PRD dead refs (sprint-plan/tasks)→WBS/EST/SCH/HND/SRS, HND version→v0.7 + แก้เลข section, เพิ่ม US-14/US-15 เข้า WBS+EST (base 80.5→86.5) propagate SCH, reconcile effort, resume window 7→30 วัน, analytics event naming, cohort size, SRS S4.4 priority, onboarding 12+medical	Product Owner
25 July 2026	0.6	เข้า SA phase — เพิ่ม SRS v0.1 (System Requirements Specification, IEEE 830-style) จาก REQ + HND. Q1/Q2 marked as blocking. Next: SAD after Q1/Q2 decided	Product Owner
25 July 2026	0.5	เพิ่ม HND (Handover Document) เตรียมส่งมอบ SA + เพิ่ม §6 System Context (External Interfaces / User Classes / Operating Environment / Data Volume / Quality Attribute Priorities) ให้ SA มี input พอสำหรับ SRS+SAD ตาม IEEE 830. อัปเดต REQ traceability เพิ่ม SA Reference column	Product Owner
25 July 2026	0.4	จัดใหม่ตาม workflow PRD→REQ→WBS→EST→SCH. เพิ่ม 4 เอกสาร standard PM: REQ (requirements catalog + traceability), WBS (hierarchical work breakdown, 100% rule), EST (effort estimates with confidence levels), SCH (schedule framework template, owner-fills). Archive sprint-plan และ sprint-tasks (redundant กับ WBS+EST+SCH)	Product Owner
25 July 2026	0.3	ถอด resource assumptions (13 สัปดาห์, 8-10 ชม./สัปดาห์, F&F 5 คน, buffer %, sprint dates) ออกจาก sprint-plan และ sprint-	Product Owner

DATE / วันที่	REV	DESCRIPTION / รายละเอียดการแก้ไข	BY / โดย
		tasks. เปลี่ยนเป็น Work Packages (WP0-WP6) ที่ไม่ผูก schedule. เก็บ effort estimates ไว้เป็น work sizing. Owner จัดการ scheduling เอง.	
25 July 2026	0.2	ปรับ layout หน้า 1 (Cover) ใช้ CSS Grid 5 rows + หน้า 2 (Doc Control) ปรับ section markers เป็น badge สีดำ, compress spacing	Product Owner
25 July 2026	0.1	ฉบับแรก (Initial release) — Cover page, doc control, body content	Product Owner

Work Breakdown Structure (WBS) — Daily Fitness Companion (Phase 1)

Purpose: Deliverable-oriented hierarchical decomposition ตาม **100% rule** (children ทั้งหมด รวมกันครอบคลุม parent 100%) — ใช้เป็นฐานของ EST และ SCH

Position in workflow: PRD → REQ → **WBS** → EST → SCH **Source:** REQ.md (requirements catalog) **Consumed by:** EST.md (per-element estimates), SCH.md (dependency + milestones)

1. WBS Numbering Convention

- Level 1: 1.0 — Project
- Level 2: 1.n — Work Package (7 packages)
- Level 3: 1.n.m — Deliverable (story)
- Level 4: 1.n.m.k — Task (implementation unit)

Each element has: **WBS Code** • **Name** • **Deliverable** • **Description** • **Predecessor** • **Owner**

2. Hierarchy Overview

```
1.0 Daily Fitness Companion – Phase 1 MVP (Project)
├── 1.1 Setup (WP0)
├── 1.2 Foundation – Onboarding & Owner Queue (WP1)
├── 1.3 Owner Plan Input (WP2)
├── 1.4 Today View MVP (WP3)
├── 1.5 Today Complete + Edit (WP4)
├── 1.6 Polish + Soft Launch (WP5)
└── 1.7 Full F&F Launch (WP6)
```

3. WBS Dictionary — Level 2 (Work Packages)

WBS Code	Name	Deliverable	Predecessor	Owner
1.1	Setup	Tech chosen, backend running, empty app deployed, prompt playbook	—	Owner
1.2	Foundation	User submits form → Owner sees queue entry	1.1	Owner
1.3	Owner Plan Input	Owner creates plan for a submission, marks ready, user notified	1.2	Owner
1.4	Today View MVP	User sees today's workout + meals in one glanceable view	1.3	Owner
1.5	Today Complete +	All 5 domains in	1.4	Owner

WBS Code	Name	Deliverable	Predecessor	Owner
	Edit	Today view; user can browse + edit		
1.6	Polish + Soft Launch	P0 shipped + polished; 1st friend onboarded	1.5	Owner
1.7	Full F&F Launch	Cohort onboarded + metrics collected + Phase 2 decision	1.6	Owner

4. WBS Dictionary — Full Detail

1.1 Setup (WP0)

WBS	Name	Deliverable
1.1.1	Tech stack decision	Decision doc — native / PWA / RN
1.1.2	Backend decision	Decision doc — Firebase / Supabase / custom
1.1.3	Data model schema	ERD + migration file, tested
1.1.4	Repo + deploy	Repo initialized, deployed empty app, CI green
1.1.5	Prompt playbook draft	v0.1 playbook + 3 test personas

Tasks under 1.1

WBS	Task	Predecessor
1.1.1.1	Compare iOS native / PWA / RN across criteria	—
1.1.1.2	Document decision + rationale	1.1.1.1
1.1.2.1	Compare Firebase / Supabase / custom	—
1.1.2.2	Document decision	1.1.2.1
1.1.3.1	Design ERD (User → Submission → Plan → Day → Item)	1.1.2
1.1.3.2	Write schema + migration	1.1.3.1
1.1.3.3	Test with mock data	1.1.3.2
1.1.4.1	Repo init + config	1.1.1
1.1.4.2	Deploy pipeline + verify	1.1.4.1
1.1.5.1	Test ChatGPT with 3 personas	—
1.1.5.2	Draft playbook v0.1	1.1.5.1

1.2 Foundation — Onboarding & Owner Queue (WP1)

WBS	Name	Deliverable
1.2.1	US-01 Onboarding submission	User submits 12-Q form, saved to backend, owner notified
1.2.2	US-04 Owner queue view	Owner sees submissions list ordered by pending
1.2.3	US-16 Disclaimer	Disclaimer text at onboarding intro + About page

Tasks under 1.2

WBS	Task
1.2.1.1	Build UI screens for 12 questions
1.2.1.2	Field validation inline
1.2.1.3	Backend POST /submissions endpoint
1.2.1.4	Frontend submit integration + loading state
1.2.1.5	Owner email notification
1.2.1.6	Pending screen post-submit
1.2.2.1	Admin route + password protect
1.2.2.2	Queue list rendering
1.2.2.3	Sort + status badge
1.2.2.4	Empty state
1.2.3.1	Disclaimer copy draft (TH+EN)
1.2.3.2	Onboarding intro screen
1.2.3.3	Static About page

1.3 Owner Plan Input (WP2)

WBS	Name	Deliverable
1.3.1	US-05 Enter workout program	Owner can input day x exercises x sets/ reps
1.3.2	US-06 Enter meals + supp + cardio + recovery	Owner input 4 additional domains
1.3.3	US-07 Mark ready + notify	Status transition + user email
1.3.4	US-03 Medical flag block	Onboarding blocks or explicit bypass

Tasks under 1.3

WBS	Task
1.3.1.1	Day picker Mon–Sun + rest toggle
1.3.1.2	Exercise input form
1.3.1.3	Add/remove/reorder exercises
1.3.1.4	Auto-save + optimistic UI
1.3.1.5	Backend <code>PUT /plans/{id}/workout</code>
1.3.2.1	Meal input (mealtime + food + portion + macros)
1.3.2.2	Supplement input (name + dose + timing)
1.3.2.3	Cardio input (type + duration + freq)
1.3.2.4	Recovery input (sleep + rest days)
1.3.2.5	Backend endpoints for 4 domains
1.3.3.1	“Mark ready” button + validation
1.3.3.2	Status transition + timestamp
1.3.3.3	User email notification
1.3.3.4	Frontend routing post-ready
1.3.4.1	Medical condition question in onboarding
1.3.4.2	Block screen + copy
1.3.4.3	Bypass with consent flow
1.3.4.4	Backend flag storage

1.4 Today View MVP (WP3)

WBS	Name	Deliverable
1.4.1	US-08 Today view workout	Glanceable workout section, offline capable
1.4.2	US-09 Today view meals	Meals section integrated in Today view

Tasks under 1.4

WBS	Task
1.4.1.1	Route + skeleton loader
1.4.1.2	Fetch today's plan (cache-first)
1.4.1.3	Day type detection
1.4.1.4	Workout card component
1.4.1.5	Offline cache implementation
1.4.1.6	Analytics events
1.4.1.7	Typography + polish
1.4.2.1	Meals card component
1.4.2.2	Optional macro display
1.4.2.3	Section ordering by time-of-day
1.4.2.4	Empty state per meal
1.4.2.5	Responsive check

1.5 Today Complete + Edit (WP4)

WBS	Name	Deliverable
1.5.1	US-10 Supp/cardio/recovery cards	Today view has all 5 domains
1.5.2	US-11 Empty + rest states	Clear state handling
1.5.3	US-12 Browse days	Week navigator UI
1.5.4	US-13 Edit item	Modal edit for all item types
1.5.5	US-14 Add/remove items	User adds/removes items within a day
1.5.6	US-15 Duplicate week	Copy this week's plan to next week (P2)

Tasks under 1.5

WBS	Task
1.5.1.1	Supplement card (sort by timing)
1.5.1.2	Cardio card
1.5.1.3	Recovery card
1.5.1.4	Section ordering integration
1.5.2.1	Pending state
1.5.2.2	Rest day state
1.5.2.3	No plan state
1.5.2.4	Copy review
1.5.3.1	Week navigator UI
1.5.3.2	Selected day state
1.5.3.3	Last viewed day persistence
1.5.3.4	Out-of-range handling
1.5.3.5	Transitions polish
1.5.4.1	Edit action per item
1.5.4.2	Edit modal — workout
1.5.4.3	Edit modal — meal
1.5.4.4	Edit modal — supp / cardio
1.5.4.5	Save + optimistic sync
1.5.4.6	Cancel + unsaved warning
1.5.5.1	Add-item action + modal (reuse edit modal)
1.5.5.2	Remove item + confirm dialog
1.5.6.1	Duplicate week action + backend copy

WBS	Task
1.5.6.2	Overwrite confirm dialog

1.6 Polish + Soft Launch (WP5)

WBS	Name	Deliverable
1.6.1	US-02 Save partial onboarding	Resume flow works
1.6.2	OPS Polish + accessibility	Typography, touch, a11y, loading, errors
1.6.3	1st user onboarding	1 friend using product with real plan

Tasks under 1.6

WBS	Task
1.6.1.1	localStorage save per step
1.6.1.2	Restore on next open
1.6.1.3	30-day expiration cleanup
1.6.1.4	Test resume flow
1.6.2.1	Typography audit
1.6.2.2	Touch target audit ($\geq 44\text{px}$)
1.6.2.3	Loading states audit
1.6.2.4	Error messages audit
1.6.2.5	VoiceOver labels
1.6.2.6	Bug backlog cleanup
1.6.3.1	Owner dogfood period
1.6.3.2	Select 1st user + consent
1.6.3.3	Onboard user (watch friction)
1.6.3.4	Generate + input plan
1.6.3.5	Follow-up

1.7 Full F&F Launch (WP6)

WBS	Name	Deliverable
1.7.1	LAUNCH additional users	Cohort onboarded (size = owner-decided)
1.7.2	OPS bug fix rolling	P0/P1 bugs fixed as reported
1.7.3	METRICS collection	PRD \$4 metrics captured
1.7.4	RETRO Phase 1 wrap	Report + Phase 2 decision

Tasks under 1.7 (per user + rolling)

WBS	Task
1.7.1.a	Invite + consent (per user)
1.7.1.b	Watch onboarding (per user)
1.7.1.c	Generate + input plan (per user)
1.7.1.d	Deliver + follow-up (per user)
1.7.2.1	Triage feedback daily
1.7.2.2	Fix P0 bugs
1.7.2.3	Fix P1 bugs (buffer)
1.7.3.1	Query analytics for §4 metrics
1.7.3.2	Manual timing interviews
1.7.3.3	Owner effort logging
1.7.3.4	Compile metric report
1.7.4.1	Self-retro
1.7.4.2	Phase 2 go/no-go analysis
1.7.4.3	Wrap report + recommendation

5. 100% Rule Verification

Every parent element's scope = sum of children's scope

Parent	Children	100% cover?
1.0 Project	1.1 to 1.7	✓ ทุก deliverable ของ Phase 1
1.1 Setup	1.1.1 to 1.1.5	✓ tech, backend, schema, deploy, playbook
1.2 Foundation	1.2.1 to 1.2.3	✓ submission flow, queue, disclaimer
1.3 Owner Input	1.3.1 to 1.3.4	✓ 5-domain input + ready + safety
1.4 Today MVP	1.4.1 to 1.4.2	✓ workout + meals (core view)
1.5 Today Complete	1.5.1 to 1.5.6	✓ 5 domains + states + browse + edit + add/remove + duplicate
1.6 Polish + Launch	1.6.1 to 1.6.3	✓ save/resume + polish + 1st user
1.7 Full Launch	1.7.1 to 1.7.4	✓ cohort + bugs + metrics + retro

6. Change Control

Adding/removing WBS elements requires: 1. Update WBS.md hierarchy 2. Update EST.md estimates (roll-up) 3. Update REQ.md traceability if new requirement 4. Regenerate PDF/DOCX with new revision